

Zepto E-commerce SQL Data Analyst Portfolio Project

This is a complete, real-world Data Analyst portfolio project based on an e-commerce inventory dataset scraped from Zepto. This project simulates real analyst workflows, from raw data exploration to business-focused data analysis.

Project Overview

The goal is to simulate how actual data analysts in the e-commerce or retail industries work behind the scenes to use SQL to:

- Set up a messy, real-world e-commerce inventory database.
- Perform Exploratory Data Analysis (EDA) to explore product categories, availability, and pricing inconsistencies.
- Implement Data Cleaning to handle null values, remove invalid entries, and convert pricing from paise to rupees.
- Write business-driven SQL queries to derive insights around pricing, inventory, stock availability, revenue, and more.

Dataset Overview

The dataset was sourced from Kaggle and was originally scraped from Zepto's official product listings. It mimics what you would typically encounter in a real-world e-commerce inventory system.

Each row represents a unique SKU (Stock Keeping Unit) for a product. Duplicate product names exist because the same product may appear multiple times in different package sizes, weights, discounts, or categories to improve visibility - exactly how real catalog data looks.

Column	Description
sku_id	Unique identifier for each product entry (Synthetic Primary Key)
name	Product name as it appears on the app
category	Product category like Fruits, Snacks, Beverages, etc.
mrp	Maximum Retail Price (originally in paise, converted to INR)
discountPercent	Discount applied on MRP
discountedSellingPrice	Final price after discount (also converted to INR)
availableQuantity	Units available in inventory
weightInGms	Product weight in grams
outOfStock	Boolean flag indicating stock availability
quantity	Number of units per package (mixed with grams for loose produce)

Project Workflow

1. Database & Table Creation

We start by creating a SQL table with appropriate data types:

```
CREATE TABLE zepto (  
  sku_id SERIAL PRIMARY KEY,  
  category VARCHAR(120),  
  name VARCHAR(150) NOT NULL,  
  mrp NUMERIC(8,2),  
  discountPercent NUMERIC(5,2),  
  availableQuantity INTEGER,  
  discountedSellingPrice NUMERIC(8,2),  
  weightInGms INTEGER,  
  outOfStock BOOLEAN,  
  quantity INTEGER  
);
```

2. Data Import

Loaded CSV using pgAdmin's import feature.

If you're not able to use the import feature, write this code instead:

```
\copy zepto(category,name,mrp,discountPercent,availableQuantity,  
            discountedSellingPrice,weightInGms,outOfStock,quantity)  
FROM 'data/zepto_v2.csv' WITH (FORMAT csv, HEADER true, DELIMITER ',', QUOTE '"', ENCODING 'UTF8');
```

Faced encoding issues (UTF-8 error), which were fixed by saving the CSV file using CSV UTF-8 format.

3. Data Exploration

- Counted the total number of records in the dataset.
- Viewed a sample of the dataset to understand structure and content.
- Checked for null values across all columns.
- Identified distinct product categories available in the dataset.
- Compared in-stock vs out-of-stock product counts.
- Detected products present multiple times, representing different SKUs.

4. Data Cleaning

- Identified and removed rows where MRP or discounted selling price was zero.
- Converted mrp and discountedSellingPrice from paise to rupees for consistency and readability.

5. Business Insights

- Found top 10 best-value products based on discount percentage.
- Identified high-MRP products that are currently out of stock.
- Estimated potential revenue for each product category.
- Filtered expensive products (MRP > INR 500) with minimal discount.
- Ranked top 5 categories offering highest average discounts.
- Calculated price per gram to identify value-for-money products.
- Grouped products based on weight into Low, Medium, and Bulk categories.
- Measured total inventory weight per product category.

How to Use This Project

- Clone the repository.
- Open zepto_SQL_data_analysis.sql.
- This file contains: Table creation, Data exploration, Data cleaning, SQL Business analysis.
- Load the dataset into pgAdmin or any other PostgreSQL client.
- Create a database and run the SQL file.
- Import the dataset (convert to UTF-8 if necessary).

Repository Setup

```
git clone https://github.com/amlanmohanty/zepto-SQL-data-analysis-project.git
cd zepto-SQL-data-analysis-project
```

Project Source References

Zepto: <https://www.zeptonow.com/>

Kaggle: https://www.kaggle.com/datasets/palvinder2006/zepto-inventory-dataset/data?select=zepto_v2.csv